

$$\Rightarrow xy + 2x + y + 2 = xy + x + 2y + 2 \Rightarrow \boxed{x = y}.$$

By Defn. 5, f is one-to-one.

Let, $f(x) = z$, where $z \in \mathbb{R}$. $\therefore x+1 = z \Rightarrow (x+1) = z \Rightarrow x(1-z) = z-1$

$$\Rightarrow \boxed{x = \frac{z-1}{1-z}}. \text{ It's easy to see that } x \in \mathbb{R} \text{ when } z \neq 1.$$

By Defn. 7, f is onto, $\therefore \forall z \in \mathbb{R}, \exists x \in \mathbb{R}$ s.t. $f(x) = z$. ($z \neq 1$)

By Defn. 8, f is a bijection. $\square$

23) Determine whether each of these functions is a bijection from $\mathbb{R}$ to $\mathbb{R}$.

a) $f(x) = 2x+1$

Solution: Let, $x, y \in \mathbb{R}$ and $f(x) = f(y)$ $\therefore 2x+1 = 2y+1 \Rightarrow 2x = 2y \Rightarrow x = y$.

By Defn. 5, f is one-to-one.

Let, $x = f(y)$ where $x \in \mathbb{R}$. $\therefore 2y+1 = x \Rightarrow y = \frac{x-1}{2}$. We know that $\frac{x-1}{2} \in \mathbb{R}$.

By Defn. 7, f is onto, since $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ s.t. $f(y) = x$.

By Defn. 8, f is a bijection. $\square$

b) $f(x) = x^2+1$

Solution: We know that $1, -1 \in \mathbb{R}$. $f(1) = 1^2+1 = 2$ $f(-1) = (-1)^2+1 = 2$

But, $1 \neq -1$. By Defn. 5, f is not one-to-one.

By Defn. 8, f is not a bijection. $\square$

c) $f(x) = x^3$

Solution: Let, $x, y \in \mathbb{R}$ s.t. $f(x) = f(y)$. $\therefore x^3 = y^3 \Rightarrow x^3 - y^3 = 0 \Rightarrow (x-y)(x^2+xy+y^2) = 0$.

$$\text{Let, } x^2+xy+y^2 = 0 \quad \therefore x = \frac{-y \pm \sqrt{y^2 - 4y^2}}{2} = \frac{-y \pm y\sqrt{-3}}{2} \notin \mathbb{R}.$$

$\therefore x-y=0 \Rightarrow x=y$. By Defn. 5, f is one-to-one.

Let, $f(x) = y$, where $y \in \mathbb{R}$. $\therefore x^3 = y \Rightarrow x = \sqrt[3]{y}$. We know that $\sqrt[3]{y} \in \mathbb{R}$.

$\therefore \forall y \in \mathbb{R}, \exists x \in \mathbb{R}$ s.t. $f(x) = y$. By Defn. 7, f is onto.

By Defn. 8, f is a bijection. $\square$